

DTSA 5020

Statistical Learning for Data Science: Regression and Classification

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Machine Learning Engineer (47.8 / 100)	34.5 / 100
SPECIALIZATION Intro to Machine Learning: Supervised Learning (Statistical Learning specialization)		

34

CURRENT AVG SCORE
across 6 job roles

59

POTENTIAL SCORE
with recommended additions

+33

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Consists of the foundational framework & application of simple and multiple linear regression and classification methods.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	47.8	68.5	+20.6	MEDIUM
Data Scientist	39.0	58.1	+19.1	MEDIUM
ML Engineer	37.5	61.9	+24.5	MEDIUM
AI Engineer	29.4	52.8	+23.4	MEDIUM
MLOps Engineer	29.0	62.5	+33.4	HIGH
Applied Scientist	24.2	52.1	+28.0	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

✓ This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	47.8 → 68.5 (+20.6 pts)
data predictive learning tools design development science models	
Data Scientist	39.0 → 58.1 (+19.1 pts)
data different problems	
ML Engineer	37.5 → 61.9 (+24.5 pts)
data learning machine models model world problems tools	
AI Engineer	29.4 → 52.8 (+23.4 pts)
data models learning model problems machine design tools	
MLOps Engineer	29.0 → 62.5 (+33.4 pts)
data model scientists learning machine tools models development	
Applied Scientist	24.2 → 52.1 (+28.0 pts)
computational problems models data learning industry design development	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.